

# TransitOps - Technical Requirements Document (TRD)

## 1. Frontend Stack

Framework: React.js (Vite)  
Language: TypeScript  
Styling: Tailwind CSS + shadcn/ui  
Routing: React Router DOM  
State Management: TanStack Query (React Query)  
Forms: React Hook Form + Zod  
HTTP Client: Axios  
Charts: Recharts  
Notifications: React Hot Toast

## 2. Backend Stack

Runtime: Node.js  
Framework: Express.js  
Language: TypeScript  
ORM: Drizzle ORM  
Validation: Zod  
Password Hashing: bcrypt  
Authentication: JWT

## 3. Database

Database: Neon PostgreSQL  
ORM: Drizzle ORM

Core Tables: Users, Roles, Vehicles, Drivers, Trips, MaintenanceLogs, FuelLogs, Expenses

## 4. Authentication Method

JWT Authentication with HTTP-only cookies, bcrypt password hashing, protected routes, and Role-Based Access Control (Fleet Manager, Dispatcher, Safety Officer, Financial Analyst).

## 5. APIs Needed

Authentication, Dashboard, Vehicles CRUD, Drivers CRUD, Trips CRUD & lifecycle, Maintenance, Fuel Logs, Expenses, Analytics, CSV Export APIs.

## 6. AI Models / Tools

No AI is required for the MVP. Future enhancements may use OpenAI GPT-5 or Google Gemini for operational insights and analytics.

## 7. Cloud / Deployment Setup

Frontend: Vercel  
Backend: Render  
Database: Neon PostgreSQL

Flow:

User → Vercel (React) → Render (Express API) → Neon PostgreSQL

## 8. Security Requirements

- JWT Authentication
- RBAC Authorization
- HTTP-only Secure Cookies
- bcrypt Password Hashing
- Zod Input Validation
- Helmet Middleware
- CORS Configuration
- Rate Limiting
- Environment Variables for Secrets
- Business Rule Validation

## 9. Performance Requirements

- API Response Time: < 300 ms
- Dashboard Load: < 2 seconds
- Lighthouse Score: 90+
- Pagination for large datasets
- Indexed database fields
- Cached API responses using TanStack Query

## Key Technical Decisions

| Decision                      | Reason                                                                    |
|-------------------------------|---------------------------------------------------------------------------|
| React + Vite                  | Fast development and optimized builds                                     |
| Express.js                    | Lightweight REST API framework                                            |
| Neon PostgreSQL + Drizzle ORM | Serverless PostgreSQL with type-safe SQL and excellent TypeScript support |
| JWT + RBAC                    | Secure authentication and authorization                                   |
| Render + Vercel               | Simple CI/CD and free-tier hosting                                        |
| Tailwind CSS + shadcn/ui      | Modern, responsive UI with rapid development                              |